Encoding the New Frontier: Adapting MEI and Verovio for Post-Tonal and Spectral Notations

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1. Introduction: Verovio and Jupyter in Computational Musicology

Verovio [8] is a prominent web-based musical score rendering tool. Its Python implementation offers a practical alternative to Music21 [3], directly processing kern, MusicXML, and MEI into SVG for seamless Jupyter/IPython integration. This circumvents external renderers (e.g., MuseScore, LilyPond), significantly simplifying Jupyter workflows [7] and enabling complete analytical pipelines in environments like Google Colab and Jupyter4NFDI. This poster presents AudioSpylt, a Jupyter-based toolbox replicating and extending functionalities of established IRCAM software (e.g., Audiosculpt, OpenMusic).

2. AudioSpylt

AudioSpylt is a Python toolbox for sound analysis, resynthesis, and symbolic/visual sound representation. Initially developed for creative explorations of DFT resynthesis, it now offers advanced analytical capabilities, particularly for spectral and acousmatic music. A key innovation is its Verovio integration for visualizing pitch structures derived from spectral analyses.

Core Features and Workflow:

- **Audio Preparation**: Interactive audio selection and editing via Plotly, including waveform navigation and precise segment isolation with cuts/fades (Fig. 1).
- **Spectral Analysis**: Frame-accurate DFT analysis with customizable resolution; harmonic peak extraction via amplitude/frequency thresholds (Fig. 2).
- **Interactive Exploration**: Plotly visualizations of TSV data (scatter, bar, line, 3D), enabling comparative spectral analyses over single frames or temporal sequences (Fig. 3).
- **Symbolic Representation**: Extended MEI encoding and enhanced Verovio rendering for accurate microtonal and graphical notation of spectral data.

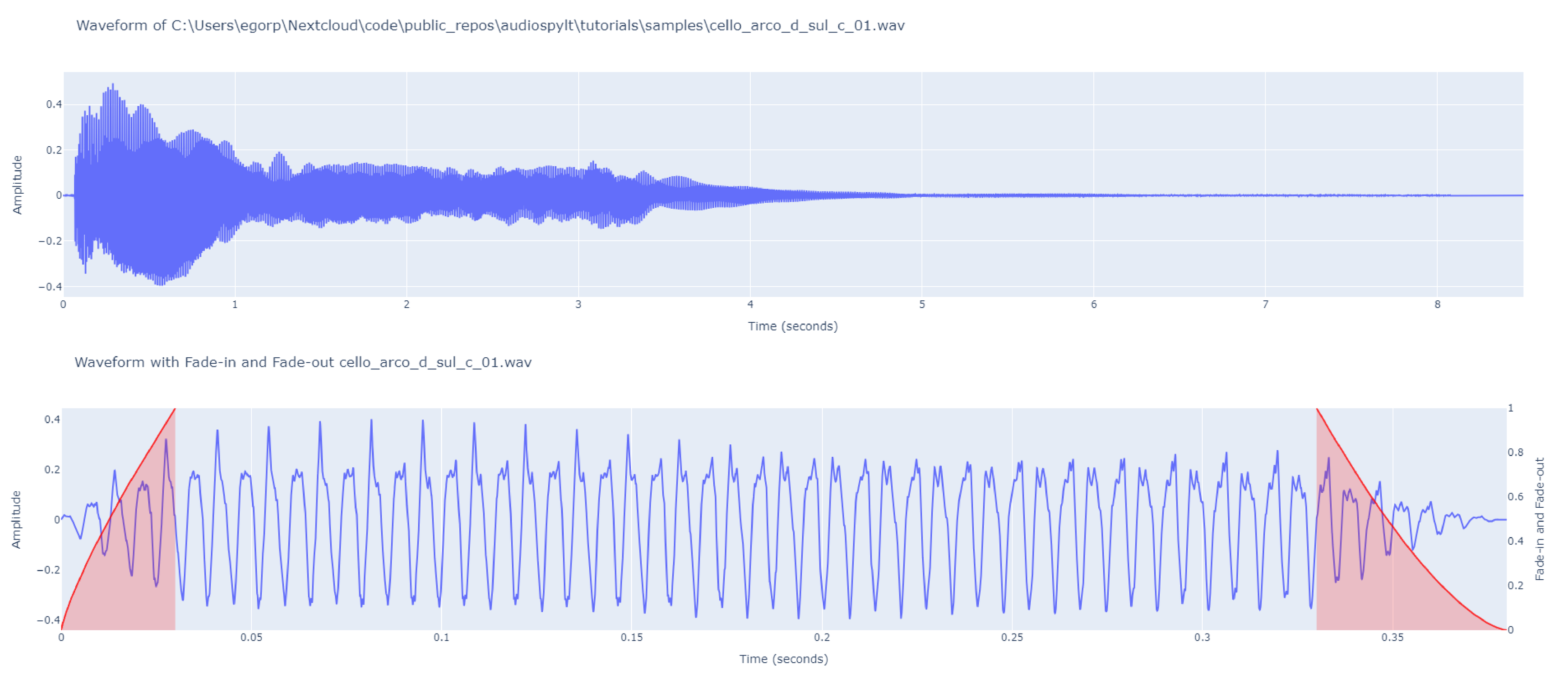

Figure 1. AudioSpylt's interactive audio preparation. Top: Navigable waveform display via Plotly. Bottom: Segment refinement using fade-in/out overlays before spectral analysis.

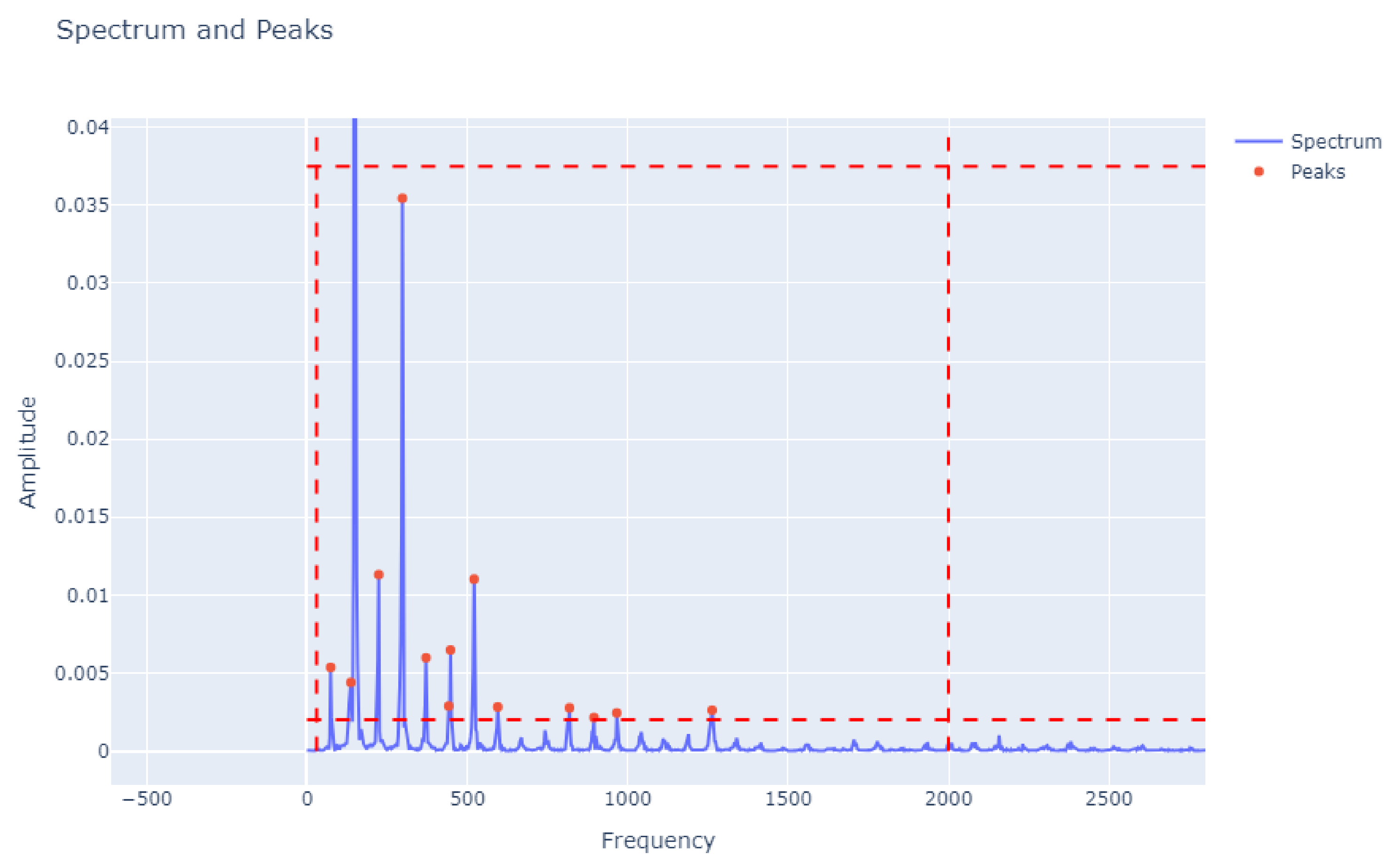

Figure 2. Threshold-based selection of harmonic peaks from DFT analysis in AudioSpylt.

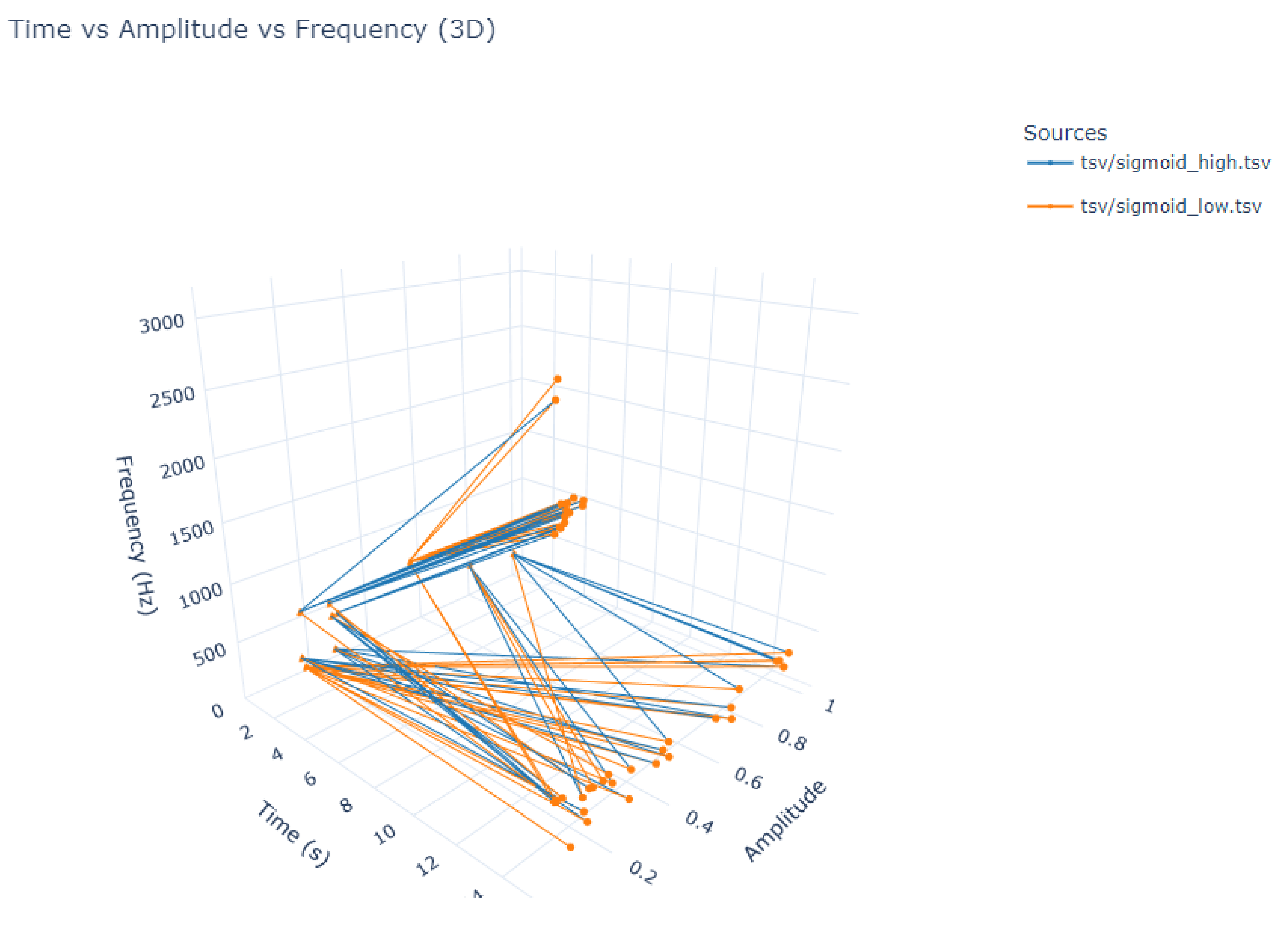

Figure 3. Comparison of spectral morphing processes across two temporal sequences.

Verovio/MEI Implementation in AudioSpylt

AudioSpylt implements symbolic frequency table representation using Verovio/MEI, featuring two microtonal notation modes:

- **Extended Accidentals**: Utilizes extended MEI accidentals (defined in `data.ACCIDENTAL.WRITTEN`, including quarter-tones in Gould and Tartini-Couper styles). These are processed by AudioSpylt to achieve 1/8-tone resolution, adhering to OpenMusic conventions (Fig. 4).
- **MIDI Cent Deviations**: Employs half-tone notation with MIDI cent deviations (via MEI's `<fing>` tag) above notes for precise microtonal specification (Fig. 5).

For temporal DFT structures, AudioSpylt uses graphical notation where chord spacing encodes temporal duration in milliseconds; this resolution is adjustable via the `measure_represents_sec` parameter. This graphical approach relies primarily on the selective visibility of MEI elements to define chord spacing according to the temporal structure. However, dense accidentals within chords can occasionally alter their intended temporal positioning (Fig. 6).

AudioSpylt exports scores as SVG (for graphics-only output) and MEI files. The produced MEI files are compatible with modern MEI-compliant editors (e.g., Verovio Editor, mei-friend).

	Frequency (Hz)	MIDI Value	Cent Deviation	Pitch Name	MEI Step	MEI Octave	MEI Alter
0	31.25	23	+21	B0	B	0	nu
1	75.00	38	+37	D2	D	2	1qs
2	127.50	48	-44	C3	C	3	1qf
3	238.75	58	+42	A#3	A	3	3qs
4	321.25	64	-45	E4	E	4	1qf
5	393.75	67	+8	G4	G	4	n
6	398.75	67	+30	G4	G	4	1qs
7	518.75	72	-15	C5	C	5	nd
8	546.25	73	-26	C#5	C	5	n
9	600.00	74	+37	D5	D	5	1qs

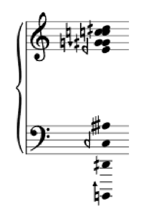

Figure 4. A frequency table represented as a chord, utilizing 1/8-tone notation.

	Frequency (Hz)	MIDI Value	Cent Deviation	Pitch Name	MEI Step	MEI Octave	MEI Alter
0	31.25	23	+21	B0	B	0	n
1	75.00	38	+37	D2	D	2	n
2	127.50	48	-44	C3	C	3	n
3	238.75	58	+42	A#3	A	3	s
4	321.25	64	-45	E4	E	4	n
5	393.75	67	+8	G4	G	4	n
6	398.75	67	+30	G4	G	4	n
7	518.75	72	-15	C5	C	5	n
8	546.25	73	-26	C#5	C	5	s
9	600.00	74	+37	D5	D	5	n

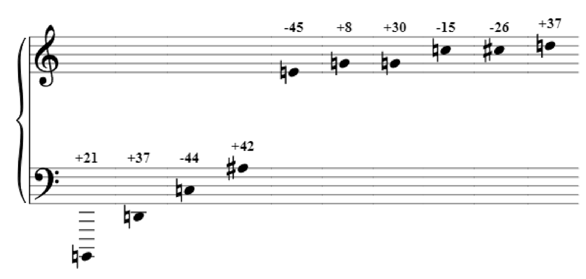

Figure 5. A frequency table represented as a sequence, employing half-tone notation with MIDI cent deviations displayed above the notes.

	Group (Measure)	Time Start	Time Stop	Num Notes	MIDI Values
0	1	0.03	0.10	18	[48, 51, 67, 72, 77, 82, 85, 85, 86, 87, 91, 9...
1	2	0.10	0.20	13	[23, 39, 47, 51, 59, 61, 69, 72, 73, 80, 81, 8...
2	3	0.20	0.40	9	[23, 38, 59, 61, 63, 67, 72, 73, 74]
3	4	0.40	0.75	1	[59]

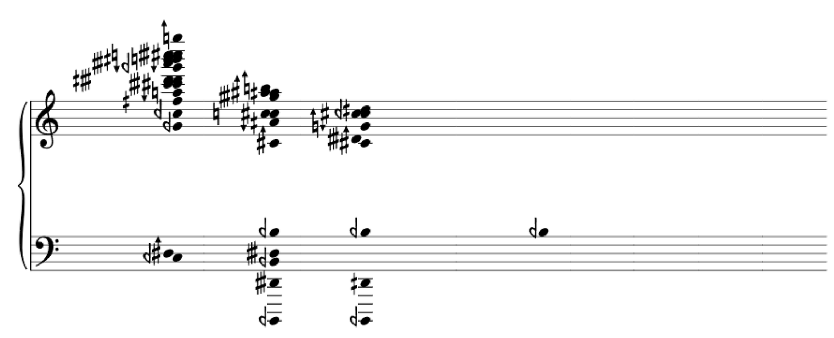

Figure 6. Temporal representation depicting frequency values extracted from several successive DFT frames. The spacing between chords encodes temporal duration in milliseconds.

3. Expanding MEI: Broader Needs for Modern Repertoire and Analysis

Implementations in AudioSpylt highlight the broader need for advanced encoding solutions in MEI. This need arises from two main developments:

- **Growing Microtonal Repertoire:** Increasing public domain availability of microtonal compositions—e.g., Charles Ives's Three Quarter-Tone Pieces (1925) and works by Russian avant-garde composers such as Arthur Lourié and Arseniy Avraamov [1].
- **Open Access Contemporary Scores:** A growing number of contemporary composers releasing scores under open licenses, facilitating analytical studies without costly licensing barriers.

Further standardizing microtonal accidental representations in MEI is essential for encoding these scores. A valuable starting point could be alignment with notations already integrated in tools like OpenMusic—encompassing not only microtonal accidentals but also basic elements for graphical notation [2] that avoid unconventional methods for displaying temporal structures (as currently explored in AudioSpylt).

Additionally, analytical musicology benefits from incorporating graphical and nested structural elements inspired by GTTM-based models [5, 6, 4]. Introducing standardized graphic elements within MEI would enhance analytical capabilities, reducing reliance on complex, compatibility-challenged SVG overlays.

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